DY PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH
Department of Computer Engineering

LABORATORY PRACTICE – III

MINI PROJECT

SURVEY REPORT ON TYPES OF BLOCKCHAINS AND THEIR REAL-TIME USE CASES

Detailed Case Study:

Blockchain in Supply Chain Management

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Abstract

Blockchain is a distributed ledger technology in which records are grouped into blocks and linked using cryptographic techniques. Copies of the ledger can be maintained by multiple authorized participants, and agreed-upon rules validate updates. This survey report introduces blockchain, describes its major types, reviews important application areas, and presents a detailed case study of blockchain in supply chain management.

Supply chains involve producers, transporters, warehouses, distributors, retailers and customers. Their records may be stored in separate systems, making it difficult to establish a consistent history of a product. A permissioned blockchain can provide shared, time-stamped records of product movements and selected events. Combined with product identifiers, sensors and reliable data-entry procedures, this can improve traceability and make unauthorized changes more detectable.

The report discusses benefits, limitations, a proposed workflow, and conditions for successful implementation. Blockchain does not automatically make incorrect data true: the accuracy of information entered, privacy, governance, cost and integration with existing systems remain important considerations.

1. Introduction to Blockchain

1.1 Definition

Blockchain is a type of distributed ledger in which transactions or records are stored in blocks linked through cryptographic hashes. Network participants follow a protocol to validate and agree on updates. Depending on its design, a network may be open to anyone or restricted to approved participants.

1.2 Basic Working

1. A participant submits a transaction or record, such as a product dispatch event.
2. The transaction is checked against the network's rules and permissions.
3. Valid transactions are grouped into a block or committed to the ledger.
4. The new block is linked to earlier records using cryptographic hashes.
5. Participants with relevant permissions can inspect the shared history and verify recorded events.

1.3 Important Features

Distributed record keeping: Copies or synchronized views of the ledger are maintained across multiple participants.

Transparency and traceability: Authorized participants can inspect a common history, depending on permissions.

Tamper evidence: Cryptographic links and consensus make unauthorized changes detectable; immutability is a design property, not an absolute guarantee.

Consensus: The network uses agreed rules to decide which updates are accepted.

Cryptographic verification: Digital signatures and hashes help verify the origin and integrity of records.

Programmability: Some blockchains support smart contracts that execute predefined rules.

1.4 Blockchain and Traditional Databases

Aspect	Traditional shared database	Blockchain
Control	Usually managed by one organization or administrator	Control distributed according to network governance
Record updates	Authorized administrators can edit records under database rules	Updates validated and appended under consensus or permission rules
Audit trail	Can be implemented; depends on configuration	Shared, cryptographically linked history is a core feature
Performance	Often efficient for high-volume centralized workloads	Depends on consensus, network design and data size
Best fit	Centralized control and flexible updates	Multi-party workflows needing shared.

2. Types of Blockchain

2.1 Public Blockchain

A public blockchain is generally open for anyone to read and, subject to protocol rules, submit transactions or participate in validation. Examples include Bitcoin and the public Ethereum network. Public networks can offer broad verifiability but may have transaction costs, variable throughput and limited privacy for on-chain data.

2.2 Private Blockchain

A private blockchain is operated by one organization or a tightly controlled administrator group. Participation and permissions are restricted. It may suit internal business processes, although its trust model is more centralized than that of a public blockchain.

2.3 Consortium or Federated Blockchain

A consortium blockchain is governed by a group of selected organizations. For example, manufacturers, logistics companies and retailers may jointly govern a shared supply-chain network. It distributes control across members while restricting access to approved participants.

2.4 Hybrid Blockchain

A hybrid blockchain combines private or permissioned records with selected public verification or disclosure. An organization may keep commercial details restricted while publishing a proof or summary that outside parties can verify.

2.5 Comparison of Blockchain Types

Type	Access	Governance	Privacy	Potential use
Public	Open	Network protocol/community	On-chain data often public	Public payments, open applications
Private	Restricted	One organization/admin group	Controlled access	Internal enterprise records
Consortium	Approved organizations	Several member organizations	Permitted participants	Inter-company supply-chain tracking
Hybrid	Mixed access	Organization(s) plus public components	Sensitive data can remain restricted	Auditable records with selective public proofs

3. Different Applications of Blockchain

3.1 Banking and Payments

Blockchain can support digital asset transfers, shared transaction records and some settlement workflows. Benefits depend on regulation, identity checks, network governance and integration.

3.2 Supply Chain Management

Producers, carriers, warehouses and retailers can record shipment events against a shared product identifier. This can help trace goods, locate delays and investigate recalls or suspected counterfeits.

3.3 Healthcare

Blockchain may provide audit trails for access to medical data or coordinate consent and integrity proofs. Sensitive clinical records are generally better kept off-chain.

3.4 Digital Identity and Credentials

Blockchain-based systems can support verifiable credentials and certificate validation. Privacy-preserving design is important; personal data should not be published indiscriminately.

3.5 Government and Public Records

A ledger can provide an auditable history for selected records, licenses or document verification. Legal validity, correction procedures and privacy must be addressed.

3.6 Education

Institutions may issue digitally verifiable certificates that employers can validate. The issuing institution remains responsible for the correctness of credentials.

3.7 Insurance and Finance

Smart contracts can automate parts of claims processing when trustworthy inputs and clear rules are available. Disputes and exceptions still require suitable processes.

3.8 Limitations Across Applications

- Incorrect input data remains incorrect even when recorded on a blockchain.
- Integration with existing systems and physical-world data sources can be expensive.
- Privacy and data-protection obligations constrain what should be recorded.
- Governance disputes, transaction costs, throughput and energy use vary by platform.
- Blockchain is not always better than a conventional database, especially where one trusted organization controls the process.

4. Detailed Survey: Blockchain in Supply Chain Management

4.1 Background and Problem Statement

A supply chain connects multiple organizations and physical locations. Product details may be recorded in separate databases, spreadsheets, paper documents and logistics systems. This fragmentation can cause inconsistent records, slow tracing, disputes about custody and difficulty identifying where a product has been delayed or compromised.

The problem addressed is how participating organizations can maintain a consistent and auditable history of selected product events while protecting commercially sensitive information.

4.2 Objectives

- Provide a shared history of product movement across participating organizations.
- Improve traceability from origin through transport, storage and sale.
- Make unauthorized changes to recorded events easier to detect.
- Support investigations, recalls and dispute resolution.
- Control access so each participant sees only information they are authorized to view.

4.3 Proposed System

A permissioned consortium blockchain is a reasonable proposed design because manufacturers, logistics providers, warehouses and retailers are known organizations that need shared records without exposing all business information publicly. Each organization operates an authorized application or node under agreed governance. Smart contracts can validate event formats and permissions.

4.4 Architecture and Workflow

Manufacturer	Logistics provider	Warehouse	Retailer	Customer / auditor
Create product ID **Record origin**	Record dispatch and custody	Record receipt and storage	Record delivery and sale	Verify permitted product history

Workflow: (1) assign a unique identifier or QR code to a product or batch; (2) record origin and production details; (3) authorized parties append dispatch, receipt, storage and delivery events; (4) the network validates and records the events; (5) a customer or auditor scans the identifier to view permitted provenance information. IoT sensors may add temperature or location data, but their accuracy and calibration must be managed separately.

4.5 Example Scenario: Tracking a Packaged Food Product

6. A producer creates batch F-2026-001 and records the production date, facility and quality-check reference.
7. A logistics company records pickup, shipment identifier and dispatch time.
8. A cold-storage warehouse records receipt and the required storage conditions.
9. A distributor records transfer to a retailer.

10. The retailer scans the batch identifier and checks the permitted history.
11. If a quality issue is reported, authorized parties use the event history to identify affected batches and locations.

This is an illustrative workflow, not a claim that a particular company uses this exact design. Real deployments vary in architecture, data-sharing rules and integration.

4.6 Data Recorded on the Blockchain

Data item	Example	Reason
Product/batch ID	F-2026-001	Links events to a traceable unit or batch
Event type	Dispatched / Received	Describes the supply-chain step
Timestamp	Recorded event time	Establishes event order
Participant ID	Authorized carrier ID	Identifies the organization reporting the event
Document reference or hash	Quality certificate reference	Supports verification without exposing all document contents
Sensor summary (optional)	Temperature range	Supports handling checks when sensor reliability is established

Sensitive contracts, personal data and large documents should usually remain in secured off-chain systems. The ledger can store a reference or cryptographic hash where suitable, but even hashes and metadata may reveal information in some contexts.

4.7 Benefits

- Improved traceability through a shared event history.
- Auditability: time-stamped records help investigate disputes and process failures.
- Counterfeit investigation support by checking provenance against authorized records.
- Recall support: affected batches and known movements can be identified more quickly when records are complete.
- Reduced reconciliation effort when organizations no longer need to compare inconsistent copies of records.
- Smart contracts can validate required fields or authorized event transitions.

4.8 Challenges and Limitations

- Data quality: false or mistaken input can be preserved; validation at entry is essential.
- Physical-to-digital link: labels can be copied, products substituted and sensors manipulated unless controls are used.
- Cost and complexity: participants need compatible processes, software, training and support.

- Confidentiality: organizations may not want to expose prices, suppliers, volumes or routes to competitors.
- Governance: members must agree on membership, access, dispute handling, upgrades and responsibility.
- Scalability and interoperability: systems must integrate with ERP, warehouse and transport platforms.
- Legal and operational fit: the technology must meet relevant laws and industry requirements.

4.9 Evaluation Criteria

A pilot should be evaluated using measurable criteria rather than assuming blockchain automatically improves performance. Suggested measures include average time to trace a batch, time to reconcile shipment records, percentage of events recorded completely, number of data discrepancies, recall identification time, system availability, cost per recorded event and user adoption.

4.10 Survey Findings

Blockchain is most relevant when several independent organizations need to write to and verify a common history, and no single participant is accepted as the sole controller. A permissioned consortium model can balance shared auditability with restricted access. However, a conventional database may be simpler and cheaper where one organization is trusted to maintain the authoritative record. Blockchain should be selected only after comparing governance, privacy, cost, performance and integration needs.

5. Conclusion

This survey introduced blockchain technology, explained public, private, consortium and hybrid blockchains, and reviewed applications across finance, supply chains, healthcare, identity, government, education and insurance. The supply-chain case study showed how a permissioned shared ledger can record product origin and movement events and provide a common audit trail to authorized participants.

Blockchain can improve traceability and coordination, but it is not a complete solution by itself. Data-entry quality, physical product identification, privacy, governance, integration and cost remain critical. A small pilot with clear performance measures is recommended before large-scale deployment.

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